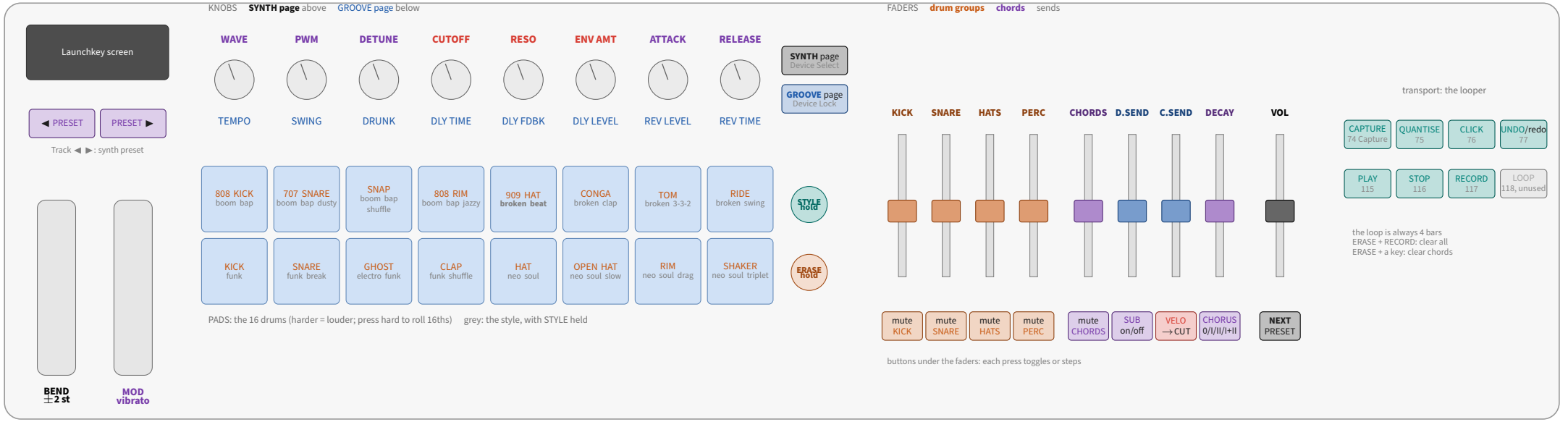


The looper

Start: hold STYLE and tap a pad: drums written in that style for the whole loop and its tempo, swing and drunk (bars 1–3 the style, bar 4 a variation). PLAY. Tap the same pad again (STYLE held) for a new variation.

Record: RECORD records what you play, on top of the loop (from stopped it starts the loop at bar 1). Press again to stop recording; the loop plays on.

Drums you play land on the 16th grid when QUANTISE is on (power-on), with the swing. **Chords** are kept exactly as played, held notes and all; the sustain pedal holds them.

Capture: play over the running loop without recording; if you like it, CAPTURE puts the last loop-length you played into the loop.

Undo takes back the last pass (recording, style, erase, clear); pressed again it redoes.

Erase: hold ERASE and tap a pad: that drum's hits leave the loop.

Groove

TEMPO 70 – 140 BPM. **SWING** as on an MPC: the first 16th of each pair takes 50 – 75 % of it. **DRUNK**: grid hits land late by a random amount, new every bar, up to a third of a 16th; never the kicks. A style sets all three (page 2); a variation keeps them. After a style, a knob takes over when turned through its value.

Chords: the synth

4 voices (a 5th note takes the oldest). Two oscillators: WAVE saw → pulse, PWM its width wobble, DETUNE brings in the second oscillator then detunes it (0 – 25 cents); SUB an octave down. 24 dB/oct low-pass: CUTOFF, RESO, ENV AMT (0 – 6 octaves of filter envelope, by velocity with VELO→CUT). ATTACK, DECAY (fader 8), RELEASE for both envelopes; the preset sets the sustains. CHORUS as the Juno's.

Mix and effects

Faders 1–4: drum groups (KICK: kicks; SNARE: snares, ghost, clap, snap; HATS: hats, ride; PERC: rims, shaker, conga, tom). Fader 5 the chords. Buttons 1–5 mute them (the loop goes on). Faders 6, 7: how much of the drums and of the chords goes to the effects: a delay in beats (DLY TIME: 1/16, 1/8, dotted 1/8, 1/4, dotted 1/4, 1/2) into the Elements reverb.

Rolls

Press a pad hard (past 3/4 of its pressure) while the loop runs: it repeats every 16th at the pressure's velocity, swung, and is recorded when recording.

Its own SD card

grains and groove each live on their own card; freak in flash when there is none. **Install**: open `groove_v1.axp`, *Upload to SD Card as Startup*: it also copies the kit to `/kit`. **Your own kit**: `/kit/NN_name.wav`, NN = 01 – 16 is the pad (any WAV: 16/24/32-bit, float, any rate, stereo mixed to mono). Without files the pads play built-in stand-ins.

Signal path

pads, loop → 16 drum voices; keys, loop → 4 synth voices → chorus → mix (sends → delay → reverb) → VOL. The loop holds up to 768 events, 4 bars.

The 16 drum styles: hold STYLE + pad

(each sets tempo, swing as MPC %, drunk 0 – 1; bars: its phrase)

pad	style	BPM	swing	drunk	bars	sounds
1	funk	100	52	0	1	909 hats, ghost notes
2	funk break	104	50	0	2	707 snare, conga
3	electro funk	112	50	0	1	808 kick, clap
4	funk shuffle	96	62	0.08	1	half time, ghosts
5	neo soul	84	62	0.50	2	808 kick, rim, shaker
6	neo soul slow	76	64	0.39	1	snap + clap
7	neo soul drag	88	60	0.71	1	snare, hats late
8	neo soul triplet	80	66	0.31	1	ride, rim, conga
9	boom bap	90	60	0.08	2	909 kick, snare
10	boom bap dusty	86	62	0.16	1	808 kick, rim
11	boom bap shuffle	92	64	0	1	clap + snare
12	boom bap jazzy	88	62	0.24	2	ride, 707 snare
13	**broken beat**	120	54	0.16	2	late snare, conga
14	broken clap	124	56	0.08	2	clap, rim, tom
15	broken 3-3-2	118	58	0.24	1	hats 3+3+3+3+2+2
16	broken swing	112	60	0.24	2	808 kick, clap, shaker

Bottom row (pads 1–8): funk, neo-soul; top row (9–16): boom-bap, broken beat. A style's parts are written with fixed strokes, optional ones (a variation keeps each half the time) and possible ones (a variation adds each a third of the time). Snare ghosts play the GHOST pad. The 4-bar loop is the phrase repeated, bar 4 a variation; tapping the pad again draws a new variation for bars 1–3 (the same in each) and a new bar 4. The patterns follow each genre's conventions; they are not copied from records.

The kit (objects/kit_bb, CC0): drum machines, dry

Bottom row: KICK, SNARE, GHOST (a softer 909 snare), RIM (909); CLAP, HAT, OPEN HAT (Michael Fischer's recordings of a real TR-808); SHAKER (727 cabasa). Top row: 808 KICK (long), 808 RIM, CONGA, TOM (Fischer's 808); 707 SNARE, SNAP (808), 909 HAT, RIDE (909). 909/707/727 sounds and the snap are Roland TR-8 recordings (*MckSamplePacks*). Closed hats cut the open hat.

Synth presets: Track ◄ / ► or NEXT PRESET

(from the parameter meanings, not yet tuned by ear)

0 juno brass one saw + sub, filter swell, chorus I · **1 prophet brass** two detuned saws, slower swell · **2 soft pad** pulse with PWM, slow, chorus I+II · **3 stab** short, resonant, chorus II · **4 pwm strings** pulse, PWM, sub · **5 dark keys** plucky, low cutoff · **6 bright brass** more detune, open · **7 reso sweep** slow resonant swell.

Where the values fall (knob or fader turned . . .)

	1/4	1/2	3/4	full
TEMPO (BPM)	87.5	105	122.5	140
SWING (MPC %)	56	62.5	69	75
DRUNK (most lateness, of a 16th)	1/12	1/6	1/4	1/3
CUTOFF (at middle C)	113 Hz	640 Hz	3.6 kHz	20 kHz
ATTACK	7 ms	55 ms	0.41 s	3 s
DECAY (fader 8)	47 ms	0.22 s	1.1 s	5 s
RELEASE	28 ms	0.16 s	0.89 s	5 s

DLY TIME in six equal zones: 1/16, 1/8, dotted 1/8, 1/4, dotted 1/4, 1/2 (of the tempo).

Something seems wrong?

No drums from the loop: PLAY? A group muted (fader buttons 1–4) or its fader down?
Built-in drum sounds: no `/kit` on this card, or still loading; upload the patch to the card again.
A chord hangs: STOP releases the loop's notes; the sustain pedal holds the keys'.
Recorded hits off the grid: QUANTISE was off (it applies to drums only).
Capture did nothing: it needs the loop running and something played since the last capture.
Lost: UNDO; or hold ERASE and press RECORD to clear the loop (undoable too).
Another patch (on the SD card): press Device Select and Device Lock together, then a top pad within 3 s: 1 freak, 2 grains, 3 groove, 4 soliton, 5 aeolian.

MIDI reference (Launchkey 49 MK3, standalone)

keys, bend, mod, pedal	ch 1: notes, pitch bend, CC 1, CC 64
knobs / faders / fader buttons	ch 1 CC 21–28 / 71–79 / 11–19
pads	ch 10 notes 36–51, velocity, poly pressure
pages (Device Select / Lock)	ch 16 CC 51 / 52 (also 106 / 107)
STYLE / ERASE (round buttons, held)	ch 16 CC 104 / 105
transport	ch 16 CC 74–77, 115–118
synth preset (track ◄ ►)	ch 16 CC 102 / 103